

- 1 What is the value of the definite integral

$$\int_{-2}^2 (3x^2 + 5 + x^3 \cos x) dx$$

- A 0
- B 16
- C 18
- D 36
- E 72

- 2 The diagram shows a rectangle of length L , width W , and area A .
Given that

$$A = \frac{3x^2 - 11x - 4}{9x^2 - 1} \quad \text{and} \quad W = \frac{x^2 - 2x - 8}{3x^2 + 14x - 5}$$

which one of the following is an expression for L in its simplest form?

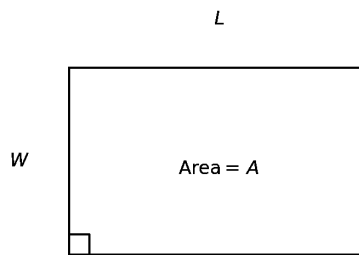


Diagram not to scale

- A $\frac{x - 5}{x + 2}$
- B $\frac{x - 5}{x - 2}$
- C $\frac{x + 2}{x + 5}$
- D $\frac{x + 5}{x - 2}$
- E $\frac{x + 5}{x + 2}$

- 3 The function $f(x)$ is continuous on the interval $0 \leq x \leq 8$. The curve $y = f(x)$ crosses the x -axis exactly once in this interval.

Given that:

$$\int_0^8 f(x) \, dx = 2$$

and the total area of the region bounded by the curve $y = f(x)$, the x -axis, and the lines $x = 0$ and $x = 8$ is 12.

What is the area of the region lying below the x -axis?

- A 2
B 5
C 6
D 7
E 10
- 4 The table shows the number of items and the mean score for three groups of data, A, B, and C.

Group	Number of items	Mean score
A	n	12
B	$2n$	18
C	k	30

When all three groups are combined, the overall mean score of all the items is 20.

What is the ratio $k : n$?

- 5 A curve has equation $y = f(x)$. The derivative of the function is given by $f'(x) = 3x^2 - 12$. The distance between the two stationary points of the curve is $4\sqrt{65}$.

What is the y-intercept of the curve?

- A -16
B 0
C 16
D 32
E Cannot be determined

- 6 The diagram shows two perpendicular straight lines, L_1 and L_2 , in the Cartesian plane.
- Line L_1 passes through the points with coordinates $(2 - k, 3k - 1)$ and $(k + 4, k + 5)$, where k is a constant.
- Line L_2 intersects the coordinate axes at $(0, 4)$ and $(6, 0)$.
- What is the value of k ?

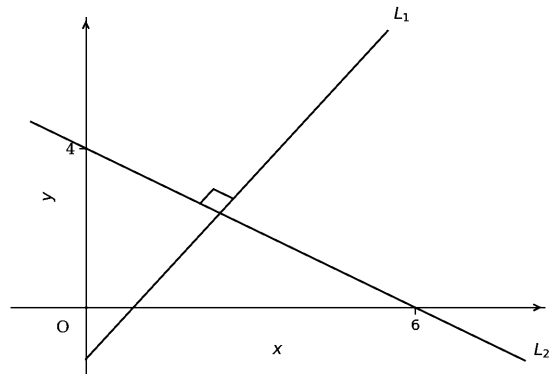


Diagram not to scale

- A -9
- B $-\frac{3}{5}$
- C $\frac{3}{5}$
- D $\frac{7}{5}$
- E $\frac{5}{3}$
- F 11
- 7 What is the sum of all real values of x that satisfy the equation $\sin(2x) = \cos(x)$ in the interval $0 \leq x \leq \pi$?
- A $\frac{\pi}{2}$
- B $\frac{2\pi}{3}$
- C π
- D $\frac{4\pi}{3}$
- E $\frac{3\pi}{2}$

- 8 A rhombus has diagonals of length $2\sqrt{5}$ cm and 4 cm. An enlargement of the rhombus has an area of $80\sqrt{5}$ cm².

What is the perimeter of the enlarged rhombus?

- A $6\sqrt{5}$ cm
- B $12\sqrt{5}$ cm
- C $24\sqrt{5}$ cm
- D $48\sqrt{5}$ cm
- E 240 cm
- F 480 cm

- 9 In triangle ABC ,

$$\angle A = 30^\circ, \quad BC = 4, \quad CA = 4\sqrt{3}.$$

What is the sum of the areas of all possible non-congruent triangles ABC satisfying these conditions?

- A $4\sqrt{3}$
- B $8\sqrt{3}$
- C $12\sqrt{3}$
- D $16\sqrt{3}$
- E $20\sqrt{3}$

- 10 The positive variables u , v , w , and x satisfy the following relationships:

- u is inversely proportional to the square root of v
- v is directly proportional to the cube of w
- w is inversely proportional to the square of x

Which of the following correctly describes the relationship between u and x ?

- A u is directly proportional to x^3
- B u is inversely proportional to x^3
- C u is directly proportional to x^6
- D u is inversely proportional to x^6
- E u is directly proportional to $x^{3/2}$
- F u is inversely proportional to $x^{3/2}$

- 11 A geometric series has a common ratio r . The value of r is the result of increasing the number 1 by $p\%$, and then decreasing the new value by $p\%$, where $p > 0$.

The sum to infinity of this series exists.

Which of the following gives the complete set of possible values for p ?

- A $0 < p < 100$
 - B $0 < p < 100\sqrt{2}$
 - C $p > 0$
 - D $p > 100\sqrt{2}$
 - E $0 < p < 200$
 - F No values of p .
- 12 The expression $(3x + 2)^6$ is expanded in ascending powers of x .
How many of the coefficients in this expansion are divisible by 6?
- A 4
 - B 5
 - C 6
 - D 7
 - E 2
- 13 The function $f(x) = ax^2 + bx + c$ has a stationary point at $x = 2$. The gradient of the curve $y = f(x)$ at $x = 0$ is 12. Given that $f(1) = 15$, what is the value of c ?
- A 0
 - B 6
 - C 9
 - D 12
 - E 24

- 14 A circle has the equation

$$x^2 + y^2 - 2x + 2y - 23 = 0$$

Find the area of the triangle bounded by the coordinate axes and the tangent to the circle at the point $(4, 3)$.

- A 6
- B 24
- C 27
- D 48
- E $\frac{49}{24}$

- 15** The function $f(x)$ is defined for $0 \leq x \leq 6$ by:

$$f(x) = \begin{cases} 11 + 2x - x^2 & \text{for } 0 \leq x \leq 3 \\ 8x - x^2 - 6 & \text{for } 3 < x \leq 6 \end{cases}$$

What is the maximum value of $f(x)$?

- A** 9
 - B** 10
 - C** 11
 - D** 12
- 16** A function f is defined by

$$f(x) = \log_3 \left(\frac{ax + b}{x} \right)$$

where a and b are constants.

Given that $f(1) = 2$ and $f(4) = 1$, find the value of x such that $f(x) = -1$.

- A** -12
 - B** -6
 - C** -4
 - D** -2
 - E** 4
- 17** How many solutions does the equation $2 \cos \left(2x + \frac{\pi}{3} \right) = 1$ have in the interval $0 \leq x \leq 2\pi$?
- A** 2
 - B** 3
 - C** 4
 - D** 5
 - E** 6
- 18** Quantities X and Y change in discrete steps $n = 0, 1, 2, \dots$. Initially, $X_0 = 20Y_0$, where $Y_0 > 0$. In each step, X decreases by 60% of its value at the start of the step, and Y decreases by 20% of its value at the start of the step.
- What is the minimum value of n such that $X_n < Y_n$?
- A** 2
 - B** 4
 - C** 5
 - D** 6

- 19** The curve with equation $y = 4^x - 10 \cdot 2^x$ intersects the horizontal line $y = c$ at two distinct points. The x -coordinates of these points are p and q , where $q > p$. Given that $q - p = 2$, what is the value of the constant c ?
- A** -25
B -24
C -16
D -4
E 16
- 20** A circle passes through the points $(2, 1)$ and $(4, 5)$ and is tangent to the x -axis. What is the largest possible radius of the circle?
- A** 1
B 2.5
C 3
D 5
E $5\sqrt{2}$
- 21** Let n be a positive odd integer. Two distinct coefficients in the expansion of $(1 + x)^n$ are chosen at random. The probability that these two coefficients are equal is $\frac{1}{23}$. What is the value of n ?
- A** 21
B 22
C 23
D 24
E 25
F 45
- 22** A sequence of three positive integers has a sum of 6 and a sum of the squares of its terms of 18. How many such distinct sequences are there?
- A** 0
B 1
C 2
D 3
E 6
F 9

- 23** The graph of $y = f(x)$ has exactly three x -intercepts:

$$-4, \quad 2, \quad 8.$$

A new function is defined by

$$g(x) = 3f(2x - 4).$$

What is the sum of the x -coordinates of the x -intercepts of $y = g(x)$?

- A** 1
B 3
C 5
D 7
E 9
F 11
- 24** A particle travels at a constant speed v from the point $(-2R, 0)$ to the point $(2R, 0)$, where $R > 0$. The particle's path is chosen to be the shortest possible route that does not enter the region $x^2 + y^2 < R^2$.

What is the minimum time for this journey?

- A** $\frac{R}{v} \left(2\sqrt{3} + \frac{\pi}{3} \right)$
B $\frac{R}{v} (\pi + 2)$
C $\frac{R}{v} \left(2\sqrt{3} + \frac{2\pi}{3} \right)$
D $\frac{R}{v} (2\sqrt{5})$
E $\frac{R}{v} (2\sqrt{3})$

- 25** A function f is defined by

$$f(x) = \frac{x^2 + 5x + 4}{x^2 + 7x + 12} \div \frac{x + 1}{3x + 9}$$

for all values of x for which the expression is defined.

What is the value of $f'(4)$?

- A** 0
B $\frac{1}{3}$
C 1
D 3
E 4
F -3

- 26** A geometric progression has first term a and common ratio r .

The first term is

$$a = \frac{1}{x+2}$$

and the sum to infinity is

$$S_{\infty} = \frac{x-1}{x^2-4}$$

What is the common ratio r ?

- A** $\frac{1}{x-1}$
- B** $\frac{x-2}{x-1}$
- C** $\frac{1}{2-x}$
- D** $\frac{2x-3}{x-1}$
- E** $-\frac{1}{x-1}$
- F** $\frac{x+1}{x+2}$

- 27 The diagram shows a square of side length $\sqrt{5} + \sqrt{2}$. A smaller square of side length $\sqrt{5} - \sqrt{2}$ is removed from one corner, leaving the shaded region.

What is the area of the shaded region?

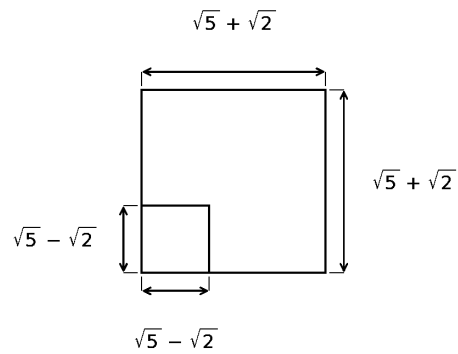


Diagram not to scale

- A 4
- B 8
- C 14
- D $2\sqrt{10}$
- E $4\sqrt{10}$
- F $7 + 2\sqrt{10}$
- G $14 - 4\sqrt{10}$
- H $14 + 4\sqrt{10}$